

FINITE DIFFERENCE METHOD FOR 2D- HEAT TRANSFER

Submitted by: Ojes Sai Pogiri

Metricola: 0054181.

AIM

To find out the Energy balance equations at different nodes for a computational domain or grid and to find Temperatures at each node.

SOFTWARES USED

Microsoft Excel

THEORY

The finite difference method is a numerical approach to solving differential equations. The fundamental equation for two-dimensional heat conduction is the two-dimensional form of the Fourier equation.

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

The finite difference method obtains an approximate solution for $\Theta(x, t)$ at a finite set of x and t . In general, Finite difference method approximate the solution to the differential equation by approximating the derivative expressions in the governing equations. This method is easy to understand but difficult to implement.

For approximating the differential increments in the temperature and space coordinates, let us consider the diagram shown below:

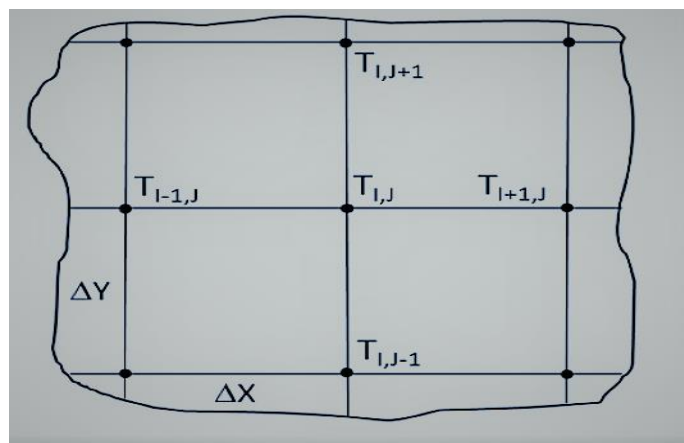


Figure 1 Temperature Coordinates of single Node

The finite difference form of the second derivatives of the temperature at node (i, j) can be determined as:

$$\frac{\partial^2 T}{\partial x^2} \Big|_{i,j} = \frac{T_{i+1,j} - 2T_{i,j} + T_{i-1,j}}{\Delta x^2}$$

$$\frac{\partial^2 T}{\partial y^2} \Big|_{i,j} = \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\Delta y^2}$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{T_{i+1,j} - 2T_{i,j} + T_{i-1,j}}{\Delta x^2} + \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\Delta y^2} = 0$$

Let us consider 2-dimensional steady state heat conduction equation with the energy generation term in the form:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{g}{k} = 0$$

Where, g = generation term

k = Thermal conductivity

COMPUTATIONAL DOMAIN OR GRID

As per the given Data:

Thermal conductivity on wall 1, $K_1=1.5\text{W/mK}$

Thermal conductivity on wall 2, $K_2=0.5\text{W/mK}$

Fluid Temperature, $T_\infty1=0^\circ\text{C}$

Wall temperature on one side, $T_\infty2=20^\circ\text{C}$

Wall temperature on other side, $T_\infty3=10^\circ\text{C}$

Convective heat transfer Coefficient, $h = 10\text{W/m}^2\text{K}$

Solar Radiation or generation term, $g = 600\text{W/m}^2$

Length of one slab $L=0.4\text{m}$

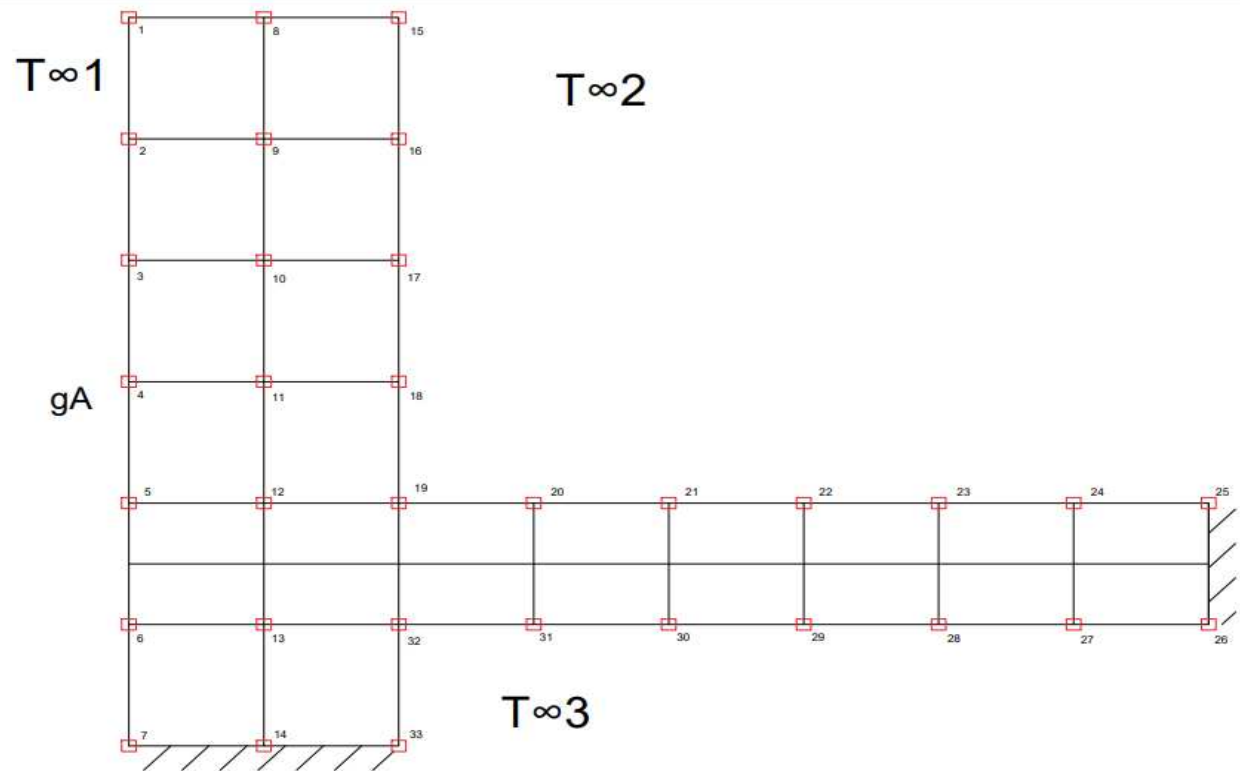


Figure 2 Domain with 26 Nodes under consideration

The term $d=L/2$

The control volume $\Delta x=d/2=L/4$

According the Energy Balance,

$$\text{Energy In} + \text{Energy Generated} = \text{Energy Out}$$

According to Fourier law

$$Q = -KA \, dt/dx$$

Grid Generation:

The actual number of nodes in the above computational grid is 33, but we consider only 25 nodes which is uniform and the distance or the thickness will be Δx .

Formulation:

🚦 Number of steps from partial to algebraic differential equations.

🚦 Mathematical:

Forward approximation and backward approximation and central difference scheme is introduced.

🚦 Physical:

This is based in energy balance.

From the Domain After considering all the 33 nodes we calculate the relation between temperatures.

When we consider one single node 1 which is taken on the left wall with thermal conductivity K_1 the temperature relation equations are taken using temperature gradient equations, similarly from the node 1 to 18 are calculated in which conduction taken place.

Exterior nodes: 1,2,3,4,5,6,7,0 ,15,16,17,18,26,27,28,29,30,31,33

Exterior corner nodes: 20,21,22,23,24,25

Interior nodes: 8,9,10,11,12,13,14

Interior corner node: 19,32

The Physical Formulation of the Finite Difference Method is given below:

The energy balance equations for all the 33 nodes are:

NODE-1	$T_1(2+(d \cdot h/K_1))-T_2-T_8 = (g+h \cdot T_{\infty 1}) \cdot d/K_1$
NODE-2	$T_1-T_2(4+(2 \cdot h \cdot d/K_1))+T_3+2T_9 = -(g+h \cdot T_{\infty 1}) \cdot 2d/K_1$
NODE-3	$T_2-T_3(4+(2 \cdot h \cdot d/K_1))+T_4+2T_{10} = -(g+h \cdot T_{\infty 1}) \cdot 2d/K_1$
NODE-4	$T_3-T_4(4+(2 \cdot h \cdot d/K_1))+T_5+2T_{11} = -(g+h \cdot T_{\infty 1}) \cdot 2d/K_1$
NODE-5	$T_4-T_5(4+(2 \cdot h \cdot d/K_1))+T_6+2T_{12} = -(g+h \cdot T_{\infty 1}) \cdot 2d/K_1$
NODE-6	$T_5-T_6(4+(2 \cdot h \cdot d/K_1))+T_7+2T_{13} = -(g+h \cdot T_{\infty 1}) \cdot 2d/K_1$
NODE-7	$T_6-T_7(2+(h \cdot d/K_1))+T_{14} = -(g+h \cdot T_{\infty 1}) \cdot d/K_1$
NODE-8	$T_1-4T_8+2T_9+T_{15} = 0$
NODE-9	$T_2+T_8-4T_9+T_{10}+T_{16} = 0$
NODE-10	$T_3+T_9-4T_{10}+T_{11}+T_{17} = 0$
NODE-11	$T_4+T_{10}-4T_{11}+T_{12}+T_{18} = 0$
NODE-12	$T_5+T_{11}-4T_{12}+T_{13}+T_{19} = 0$
NODE-13	$T_6+T_{12}-4T_{13}+T_{14}+T_{32} = 0$
NODE-14	$T_7+2T_{13}-4T_{14}+T_{33} = 0$
NODE-15	$T_8+T_{15}(2+h \cdot d/K_1)-T_{16} = -(h \cdot d \cdot T_{\infty 2})/K_1$
NODE-16	$2T_9+T_{15}-T_{16}(4+(2 \cdot h \cdot d/K_1))+T_{17} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_1$
NODE-17	$2T_{10}+T_{16}-T_{17}(4+2 \cdot h \cdot d/K_1)+T_{18} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_1$
NODE-18	$2T_{11}+T_{17}-T_{18}(4+2 \cdot h \cdot d/K_1)+T_{19} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_1$
NODE-19	$T_{12}(2 \cdot K_1)+T_{18}(K_1)-T_{19}((2 \cdot h \cdot d+4 \cdot K_1+2 \cdot K_2))+T_{20}(K_2)+T_{32}(K_1+K_2) = -(2 \cdot h \cdot d \cdot T_{\infty 2})$
NODE-20	$T_{19}-T_{20}(4+2 \cdot h \cdot d/K_2)+T_{21}+2T_{31} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_2$
NODE-21	$T_{20}-T_{21}(4+2 \cdot h \cdot d/K_2)+T_{22}+2T_{30} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_2$
NODE-22	$T_{21}-T_{22}(4+2 \cdot h \cdot d/K_2)+T_{23}+2T_{29} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_2$
NODE-23	$T_{22}-T_{23}(4+2 \cdot h \cdot d/K_2)+T_{24}+2T_{28} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_2$
NODE-24	$T_{23}-T_{24}(4+2 \cdot h \cdot d/K_2)+T_{25}+2T_{27} = -(2 \cdot h \cdot d \cdot T_{\infty 2})/K_2$
NODE-25	$T_{24}-T_{25}(2+h \cdot d/K_2)+T_{26} = -(h \cdot d \cdot T_{\infty 2})/K_2$
NODE-26	$T_{25}-T_{26}(2+h \cdot d/K_2)+T_{27} = -(h \cdot d \cdot T_{\infty 3})/K_2$
NODE-27	$2T_{24}+T_{26}-T_{27}(4+2 \cdot h \cdot d/K_2)+T_{28} = -(2 \cdot h \cdot d \cdot T_{\infty 3})/K_2$
NODE-28	$2T_{23}+T_{27}-T_{28}(4+2 \cdot h \cdot d/K_2)+T_{29} = -(2 \cdot h \cdot d \cdot T_{\infty 3})/K_2$
NODE-29	$2T_{22}+T_{28}-T_{29}(4+2 \cdot h \cdot d/K_2)+T_{30} = -(2 \cdot h \cdot d \cdot T_{\infty 3})/K_2$
NODE-30	$2T_{21}+T_{29}-T_{30}(4+2 \cdot h \cdot d/K_2)+T_{31} = -(2 \cdot h \cdot d \cdot T_{\infty 3})/K_2$
NODE-31	$2T_{20}+T_{30}-T_{31}(4+2 \cdot h \cdot d/K_2)+T_{32} = -(2 \cdot h \cdot d \cdot T_{\infty 3})/K_2$
NODE-32	$T_{13}(2 \cdot K_1)+T_{19}(K_1+K_2)+T_{31}(K_2)-T_{32}((2 \cdot h \cdot d+4 \cdot K_1+2 \cdot K_2))+T_{33}(K_1) = -(2 \cdot h \cdot d \cdot T_{\infty 2})$
NODE-33	$T_{14}+T_{32}-T_{33}(2+(h \cdot d/K_1)) = -(h \cdot d \cdot T_{\infty 3})/K_1$

When 19th Node is considered in the grid i.e., the control volume, we know that there is conduction and convection taking place, so the coefficient of the temperature from both the directions using forward and backward approximation the thermal conductivity will be the average value.

All the temperature relations are taken in to the excel software and final temperatures are known.

Plotting in MS Excel

INPUT DATA	Value	Units
Thermal conductivity K1	1.5	w/mk
Thermal conductivity K2	0.5	w/mk
Temperature at, $T_{\infty 1}$	0	$^{\circ}\text{C}$
Temperature at, $T_{\infty 2}$	20	$^{\circ}\text{C}$
Temperature at, $T_{\infty 3}$	10	$^{\circ}\text{C}$
Convection co-efficient, h	10	$\text{w/m}^2\text{k}$
Heat rate per unit volume , generation term, g	600	w/m^2
Length, L	0.4	m
3L	1.20	m
2L	0.8	m
L/2	0.2	m
L/4	0.1	m
d	0.2	m
h*d	2	

Then we define a matrix of size 26x26 as we have got solutions for 26 nodes and will plot on the excel sheet by defining the coefficient of temperatures from T_0 to T_{25} .

Table 1

	T0	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	T19	T20	T21	T22	T23	T24	T25
NODE 0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1
NODE 1	0	3.333333	-1	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NODE 2	0	1	-6.66667	1	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NODE 3	0	0	1	-6.66667	1	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NODE 4	0	0	0	1	-6.66667	1	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NODE 5	0	0	0	0	1	-6.66667	1	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
NODE 6	0	0	0	0	0	1	-6.66667	1	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0
NODE 7	0	0	0	0	0	0	1	-3.33333	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
NODE 8	0	1	0	0	0	0	0	0	-4	2	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
NODE 9	0	0	1	0	0	0	0	0	1	-4	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
NODE 10	0	0	0	1	0	0	0	0	0	1	-4	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0
NODE 11	0	0	0	0	1	0	0	0	0	0	1	-4	1	0	0	0	0	0	1	0	0	0	0	0	0	0
NODE 12	0	0	0	0	0	1	0	0	0	0	0	1	-4	1	0	0	0	0	0	1	0	0	0	0	0	0
NODE 13	1	0	0	0	0	0	1	0	0	0	0	0	1	-4	1	0	0	0	0	0	0	0	0	0	0	0
NODE 14	1	0	0	0	0	0	0	1	0	0	0	0	0	2	-4	0	0	0	0	0	0	0	0	0	0	0
NODE 15	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	3.333333	-1	0	0	0	0	0	0	0	0	0
NODE 16	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	-6.66667	1	0	0	0	0	0	0	0	0
NODE 17	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	-6.66667	1	0	0	0	0	0	0	0
NODE 18	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	-6.66667	1	0	0	0	0	0	0
NODE 19	-1	0	0	0	0	0	0	0	0	0	0	0	-3	0	0	0	0	0	-1.5	11	-0.5	0	0	0	0	0
NODE 20	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	-12	1	0	0	0	0
NODE 21	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	-12	1	0	0	0
NODE 22	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	-12	1	0	0
NODE 23	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	-12	1	0
NODE 24	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	-12	1
NODE 25	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	-6

After defining the temperature coefficients at each node in a matrix form, the inverse of the matrix is plotted in the excel sheet.

Table 2

INVERSE MATRIX OF TABLE 1																									
0.208333	2.97E-10	-4.23417E-10	-8.79608E-10	-1.82663E-09	-2.94505E-09	-1.42323E-09	-9.17708E-10	-5.65487E-10	-1.6465E-09	-3.614E-09	-8.35209E-09	-1.63830E-08	-5.62546E-09	-1.63579E-09	3.19E-10	-4.97097E-10	-1.3487E-09	-4.88025E-09	1.52E-08	-1.46181E-07	-1.74656E-06	-2.08125E-05	-0.000248004	-0.00295523	-0.03521
0.001165	0.365248	-0.070882572	-0.021665838	-0.007142656	-0.002553799	-0.000946632	-0.00563371	-0.13861779	-0.1386363	-0.0584637	-0.02388071	-0.00939393	-0.03137711	-0.0009427	0.050762	-0.030546473	-0.01424434	-0.00595212	0.001359	-5.70215E-05	-4.79495E-06	-5.17949E-06	-4.00389E-06	-1.65267E-05	-0.0002
0.001697	0.070833	-0.15945677	-0.04285118	-0.01209574	-0.003981505	-0.001421	-0.000838187	-0.069138489	-0.16789411	-0.081005	-0.033671468	-0.013026651	-0.004653641	-0.001372957	0.030547	-0.0325055	-0.01826203	-0.00826411	0.001939	-8.13826E-05	-6.84377E-06	-7.42629E-07	-2.06770E-06	-2.40708E-05	-0.00029
0.03746	0.021666	-0.042985118	-0.183881308	-0.038787414	-0.010792949	-0.00346622	-0.001979505	-0.029234342	-0.0810031	-0.143020	-0.070575173	-0.006789356	-0.010496014	-0.00112213	0.014251	-0.018269729	-0.02653666	-0.015479326	0.004029	-0.00016905	-1.42191E-05	-1.56483E-06	-4.53807E-06	-5.31416E-05	-0.00063
0.008934	0.007143	-0.01209574	-0.038787414	-0.182525637	-0.038976169	-0.00559542	-0.005524421	-0.011713141	-0.03370802	-0.0706281	-0.138073996	-0.066755946	-0.025838655	-0.007851594	0.006002	-0.008293146	-0.01557774	-0.024930342	0.000367	-0.00051088	-2.95377E-05	-3.36508E-06	-1.08432E-05	-0.000126753	-0.00051
0.002485	0.002554	-0.003891505	-0.010792949	-0.038787414	-0.181813229	-0.03945454	-0.017308371	-0.045311539	-0.01213662	-0.028953	-0.067234949	-0.01336567	-0.063294761	-0.020305703	0.002374	-0.003382911	-0.0069819	-0.014167706	0.001349	-0.000560508	-4.76742E-05	-6.03284E-06	-2.47138E-05	-0.00029605	-0.000346
0.040584	0.000394	-0.001421	-0.00346622	-0.010559542	-0.003945345	-0.0017041861	-0.0167433	-0.001724441	-0.00504348	-0.0112609	-0.027455272	-0.065494961	-0.135815957	-0.015875272	0.000911	-0.001311058	-0.00278626	-0.006003057	0.006519	-0.000273576	-2.32964E-05	-5.98087E-06	-4.8474E-05	-0.000575707	-0.00066
0.046536	0.000563	-0.000838187	-0.001979505	-0.005524421	-0.017928372	-0.01761433	-0.35594836	-0.001039716	-0.00304504	-0.0068341	-0.016921597	-0.042323727	-0.103905659	-0.114975124	0.000055	-0.000795128	-0.00170536	-0.003738862	0.004004	-0.000176418	-1.51926E-05	-5.89317E-06	-5.55255E-05	-0.0006413	-0.00787
0.002186	0.138662	-0.069518469	-0.029234342	-0.01713141	-0.004531159	-0.001724441	-0.001389716	-0.392807461	-0.294227	-0.113864	-0.044322102	-0.016770147	-0.005925396	-0.00174278	0.138661	-0.069516076	-0.02921911	-0.01161399	0.002359	-0.000106889	-9.1394E-07	-9.83854E-07	-2.66804E-06	-3.10183E-05	-0.00037
0.0032	0.069318	-0.083947656	-0.04051054	-0.01685401	-0.006598308	-0.00252174	-0.001522519	-0.147113502	-0.44981872	-0.1693666	-0.065251604	-0.024612972	-0.008698075	-0.002553323	0.069317	-0.083943567	-0.04048794	-0.016709398	0.003771	-0.000158232	-1.93055E-05	-1.43404E-06	-3.90289E-06	-4.54018E-05	-0.00054
0.00717	0.029232	-0.040507485	-0.017551427	-0.035314039	-0.014487638	-0.006530463	-0.003417046	-0.056932001	-0.1693663	-0.401188	-0.149377844	-0.05706553	-0.019621715	-0.00579369	0.02923	-0.040499772	-0.07150232	-0.034949375	0.008277	-0.000347314	-2.92063E-05	-3.16226E-06	-8.74076E-06	-0.000101727	-0.00121
0.01766	0.011699	-0.006835734	-0.025287586	-0.069036988	-0.033617475	-0.013727635	-0.008460797	-0.022161051	-0.0652516	-0.1493778	-0.391341594	-0.141351864	-0.049439295	-0.014475023	0.011699	-0.018817501	-0.0351715	-0.068281336	0.019131	-0.000802732	-6.75152E-05	-7.41808E-06	-2.59495E-05	-0.000250636	-0.00039
0.04523	0.00447	-0.006513325	-0.014339678	-0.003377974	-0.066828285	-0.032717479	-0.021161862	-0.008385073	-0.02461297	-0.0557066	-0.141351864	-0.379426547	-0.130126382	-0.037822061	0.004458	-0.006474268	-0.01409101	-0.031793254	0.019131	-0.001574527	-0.000132497	-1.54363E-05	-5.27480E-05	-0.00061455	-0.00736
0.101761	0.001587	-0.002326821	-0.005234507	-0.012948178	-0.031647382	-0.067907979	-0.051952829	-0.002962698	-0.00869078	-0.0196217	-0.049439299	-0.130126389	-0.369119946	-0.105368119	0.001573	-0.002281174	-0.00494389	-0.011056367	0.012884	-0.000540683	-4.62214E-05	-1.39732E-05	-0.000221457	-0.01443515	-0.0172
0.114603	0.000934	-0.001372957	-0.00311223	-0.007855195	-0.002305785	-0.051875772	-0.114975124	-0.001741278	-0.0510665	-0.0115194	-0.02895051	-0.07564413	-0.210536238	-0.331377841	0.000924	-0.001339369	-0.00289829	-0.00646315	0.007493	-0.000314402	-2.74355E-05	-1.3663E-05	-0.000136611	-0.01625668	-0.01937
0.00118	0.050762	-0.030547191	-0.01425122	-0.000301886	-0.002374221	-0.0001065	-0.000550453	-0.138661061	-0.1386424	-0.0584941	-0.02387131	-0.008915605	-0.03146523	-0.000941994	0.365248	-0.078830680	-0.02165632	-0.007085042	0.001046	-6.12699E-05	-5.15161E-06	-5.49385E-07	-4.4102E-06	-1.67420E-05	-0.0002
0.001747	0.030546	-0.0325055	-0.018269729	-0.008293146	-0.003382911	-0.001311058	-0.009316076	-0.16788713	-0.080995	-0.033635001	-0.012948536	-0.004562347	-0.001339369	0.078831	-0.193452819	-0.04296829	-0.020002817	0.002277	-9.5444E-05	-8.03262E-06	-8.47431E-07	-2.13655E-06	-2.47912E-05	-0.0003	
0.004069	0.014244	-0.018262032	-0.026526661	-0.01557775	-0.006981902	-0.00716256	-0.001705362	-0.02921911	-0.00897588	-0.0430406	-0.070343004	-0.028182022	-0.00988776	-0.00289828	0.021656	-0.042968292	-0.1838218	-0.039514941	0.006178	-0.000259225	-2.17881E-05	-2.23077E-06	-4.99686E-06	-5.77281E-05	-0.00069
0.011037	0.003952	-0.00826411	-0.015477826	-0.024930342	-0.014167786	-0.000380856	-0.003738861	-0.01161399	-0.0394188	-0.0699887	-0.13652671	-0.065318057	-0.02211278	-0.00645417	0.070085	-0.012002817	-0.03951494	-0.18344371	0.022354	-0.00093799	-7.88089E-05	-7.70851E-06	-1.36931E-05	-0.000156609	-0.00187
0.03418	0.002038	-0.00209342	-0.00604987	-0.012550551	-0.002023555	-0.00977808	-0.00363451	-0.00388538	-0.0131289	-0.0249313	-0.057391615	-0.11237904	-0.038651825	-0.01239319	0.02019	-0.00341546	-0.00296675	-0.03531528	0.10459	-0.00488636	-0.0003568	-3.43218E-05	-4.3302E-05	-0.000485062	-0.00578
0.041038	0.000171	-0.00024444	-0.00057188	-0.00152345	-0.001588131	-0.000208493	-0.000529155	-0.000236662	-0.0004938	-0.0020639	-0.004816318	-0.00944868	-0.00324671	-0.00049236	0.000184	-0.000286628	-0.0007777	-0.002813974	0.000777	-0.000286359	-0.0007384	-0.00059771	-9.86717E-05	-0.00056346	-0.00684
0.041614	1.44E-05	-2.04807E-05	-4.25633E-05	-8.83809E-05	0.000142508	-6.88888E-05	-4.4007E-05	-2.73633E-05	-7.9673E-05	-0.0001749	-0.000404188	-0.00072796	-0.00027221	-7.91544E-05	1.54E-05	-2.4054E-05	-6.5326E-05	-0.00023815	0.000737	-0.00077533	-0.00045147	-0.000796578	-0.000644765	-0.000640593	-0.00704
0.041662	1.2E-06	-1.71955E-06	-3.57209E-06	-7.41795E-06	-1.9598E-05	-5.77975E-05	-3.72801E-06	-2.29644E-06	-6.6864E-06	-1.480E-05	-3.39211E-05	-6.65346E-05	-2.2845E-05	-6.64296E-06	1.29E-06	-2.01871E-06	-5.4771E-06	-1.98107E-05	6.18E-05	-0.00059964	-0.00070977	-0.000451906	-0.000454929	-0.00119384	-0.000714
0.041666	1.01E-07	-1.44835E-07	-2.9946E-07	-6.22821E-07	-1.00426E-06	-4.8323E-07	-1.23980E-07	-1.92831E-07	-5.6146E-07	-1.231E-06	-2.84034E-06	-5.58808E-06	-1.91808E-06	-5.74039E-07	1.09E-07	-1.6951E-07	-4.5991E-07	-1.66419E-06	5.19E-06	-4.89476E-06	-0.00055576	-0.000705706	-0.000464617	-0.00073594	-0.00073
0.041667	8.4E-09	-1.22778E-08	-2.55806E-08	-5.29723E-08	-8.45056E-08	-4.2533E-08	-1.26913E-08	-2.6393E-08	-1.63991E-07	-4.7749E-08	-1.046E-07	-2.5139E-07	-6.13130E-07	-2.57808E-08	1.29E-08	-1.44158E-08	-3.9112E-08	-1.51527E-07	4.41E-07	-4.29827E-08	-6.56052E-05	-0.000039593	-0.0007392104	-0.00573048	-0.002123
0.041667	1.82E-08	-2.11708E-08	-4.38004E-08	-9.13316E-08	-1.47535E-08	-1.1167E-08	-4.58045E-08	-1.26991E-07	-8.2325E-08	-1.046E-07	-8.17645E-08	-1.919E-08	-2.82739E-08	-8.17896E-08	1.54E-08	-2.24848E-08	-6.9142E-08	-2.44012E-08	7.61E-08	-7.30940E-07	-8.37278E-06	-0.000104063	-0.0001240108	-0.01477615	-0.1787

RHS
1132
444
-888
-888
-888
-888
-888
-444
0
0
0
0
0
0
0
390.6667
-781.333
-781.333
-781.333
1172
-2344
-2344
-2344
-2344
-2344
-1172

	TEMP(K)	TEMP(°C)
NODE 0	284.6667	11.66666
NODE 1	323.92	50.91997
NODE 2	323.6999	50.69992
NODE 3	322.8969	49.89691
NODE 4	321.1541	48.15407
NODE 5	318.6558	45.65581
NODE 6	319.1489	46.14895
NODE 7	319.37	46.36997
NODE 8	312.0333	39.03331
NODE 9	311.5913	38.59128
NODE 10	309.8961	36.89605
NODE 11	305.7372	32.73721
NODE 12	298.0345	25.03451
NODE 13	300.8169	27.81694
NODE 14	301.4176	28.41763
NODE 15	301.0307	28.03074
NODE 16	300.7358	27.73582
NODE 17	299.3588	26.35882
NODE 18	293.8642	20.8642
NODE 19	266.9281	-6.0719
NODE 20	289.2852	16.28524
NODE 21	291.1615	18.16146
NODE 22	291.3189	18.31891
NODE 23	291.3321	18.33212
NODE 24	291.3332	18.33323
NODE 25	291.3333	18.33332

The RHS terms will be the generation term and the diameter i.e. $d=L/2$ will be depending on the control volume. In this application the length of control volume is taken as $d/2$ and the thermal conductivity values as from the equations are being calculated.

The temperature at each node is found out by multiplying the inverse matrix coefficients with the RHS values.

RESULTS

The temperature at each node has been successfully calculated using the Finite Difference Method and the results are shown below in the table.

REFERENCES

[1] Professor D. M. Causon, Professor C. G. Mingham, Introductory Finite Difference Methods for PDEs, ISBN 978-87-7681-642-1.

[2] Pradip Dutta/IISc, Bangalore, Heat and Mass Transfer, V1/18.05.2004/1